

## Lab Overview – As of 03/07/2025

### ✓ Project Overview: Automated Windows Deployment & Domain Integration Lab

#### 🧰 Core Technologies Used:

- **Windows Server (2019/2022)**
  - **Active Directory (AD DS)**
  - **DNS**
  - **DHCP with PXE booting**
  - **Windows Deployment Services (WDS)**
  - **Hyper-V with NAT + DHCP isolation**
  - **Unattended Windows installation via XML**
  - **Client OS: Windows 10 Pro**
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## 🏠 1. Lab Environment Architecture

### 💻 Domain Controller (DC1)

- Installed **Active Directory, DNS, DHCP, and WDS**
- Created and promoted domain: stellar.local
- Configured DHCP scope and PXE boot options
- Isolated with NAT/DHCP for no interference with real LAN

### 💻 Windows Deployment Server

- Same as DC1 — WDS installed and configured here
  - Imported and configured boot.wim and install.wim (Windows 10)
  - Enabled **Unattended Installation Mode**
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## ⚙️ 2. PXE Boot Setup

- Configured Hyper-V VM to boot via PXE

- Client VM boots into WDS and loads boot.wim
  - Manual test confirmed WDS was **functioning end-to-end**
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### 3. Unattended Installation Attempts

- Used a **custom-generated unattend.xml** to:
  - Automate Windows setup (skip UI)
  - Attempt automatic domain join
  - Assign a static computer name (Stellar)
  - Configure locale, time zone, and auto-accept EULA

#### Issues:

- OOBE (Out-of-Box Experience) wasn't fully skipped
- Still prompted to create a local user
- Domain join and computer rename failed

 Root Cause: The unattend.xml was injected **only at the install image phase**, not fully utilized during specialize or OOBE phases. Need to refine with **WSIM (Windows System Image Manager)** later.

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### 4. Active Directory Work

- Created a domain user: **John Doe** (jdoe)
  - Assigned to Domain Users
  - Tried logging into client VM with this domain account
  - Encountered error: "Remote Desktop Services permission required" due to using **Hyper-V Enhanced Session**, which mimics RDP
    - Resolved by planning to add the user to **Remote Desktop Users** group
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### 5. Network Isolation Achievements

- Configured NAT networking inside Hyper-V

- Ensured **lab has DHCP + internet** but doesn't conflict with host/real network
  - Fully isolated domain and IP scope
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## 6. Outcomes & Reflection

You successfully:

- Set up **WDS from scratch**
  - Built and configured **DHCP, AD, and DNS**
  - Configured **PXE boot** and served Windows 10 to clients
  - Booted and installed Windows OS from PXE image
  - Understood and began testing **unattended XML automation**
  - Created working domain users
  - Built a **virtual network that's isolated but functional**
  - Documented and troubleshoot real-world enterprise-grade setup
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## To-Do / Future Improvements

- Refine unattend.xml using **WSIM**
  - Split roles into **OUs** (e.g., Students, Staff, Devices)
  - Practice **Group Policy** (disable control panel, enforce wallpaper, etc.)
  - Add scripts for **auto domain join and rename**
  - Build out a **fake school or company** IT infrastructure
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## How This Helps Your CV

You now have valid experience with:

- PXE boot and Windows deployment
- WDS setup and troubleshooting
- Active Directory + user provisioning

- DHCP + DNS configuration
- XML scripting and Windows setup automation
- Virtual lab management (Hyper-V, NAT)
- Troubleshooting OOBE and domain join issues